Excretion, isolation and structure of a new phenolic constituent of female urine.

The regular occurrence of a peak due to an unidentified substance (X) in the gas chromatographic traces obtained from phenolic extracts of urine from human pregnant and non-pregnant females has been reported. The biphasic excretion of X with maxima in the luteal phase of the ovulatory cycle and relatively high levels in the first trimester of pregnancy were noteworthy and suggested that the substance may have a biological significance. Close similarities between the excretory pattern, the chemical and chromatographic properties of X and of those of the known phenolic steroids suggested initially that this compound was steroidal in nature. The same, or a similar, substance seems to be excreted in the vervet monkey (Cercopithecus aethiops pygerythrus). We now report the excretory pattern of X in more detail, the isolation of the pure compound from pooled pregnancy urine and the chemical structure. The structure determined by mass spectrometry, IR spectroscopy and NMR spectrometry is: trans-(+/-)-3,4-bis[(3-hydroxyphenyl)methyl]dihydro-2-(3H)-furanone (HPMF) and was confirmed by synthesis.